

DSA0601 • DATA HANDLING AND VISUALIZATION

Course Details	Information
Department	AIML
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Programming / Visualization Tools	R / Tableau
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Aim

To analyze website traffic data using a line chart, bar chart and table, and to build an interactive Tableau dashboard for exploring website traffic.

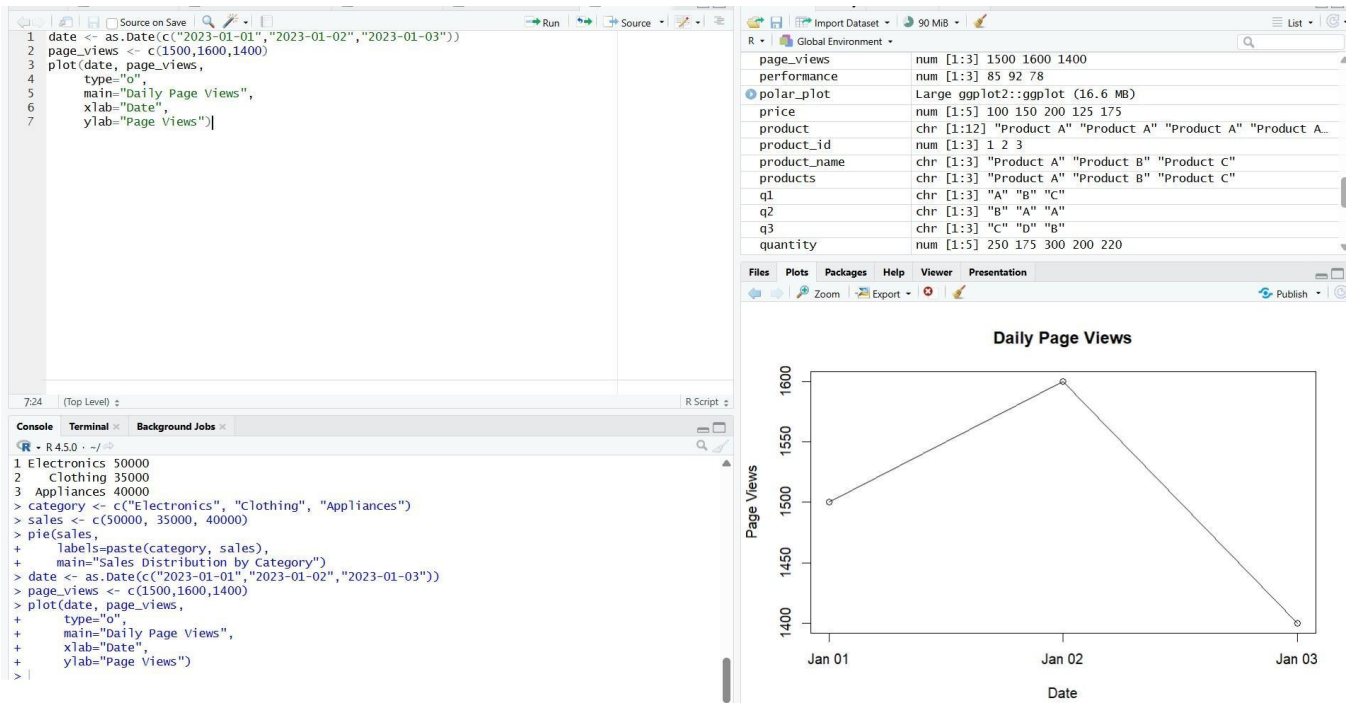
Dataset: Website Traffic

Date	Page Views	Click-through Rate
2023-01-01	1500	2.3%
2023-01-02	1600	2.7%
2023-01-03	1400	2.0%

Question 1: Line Chart of Daily Page Views

Question: Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.

Aim: To visualize the trend in daily page views over time using a line chart with labelled axes and a title.

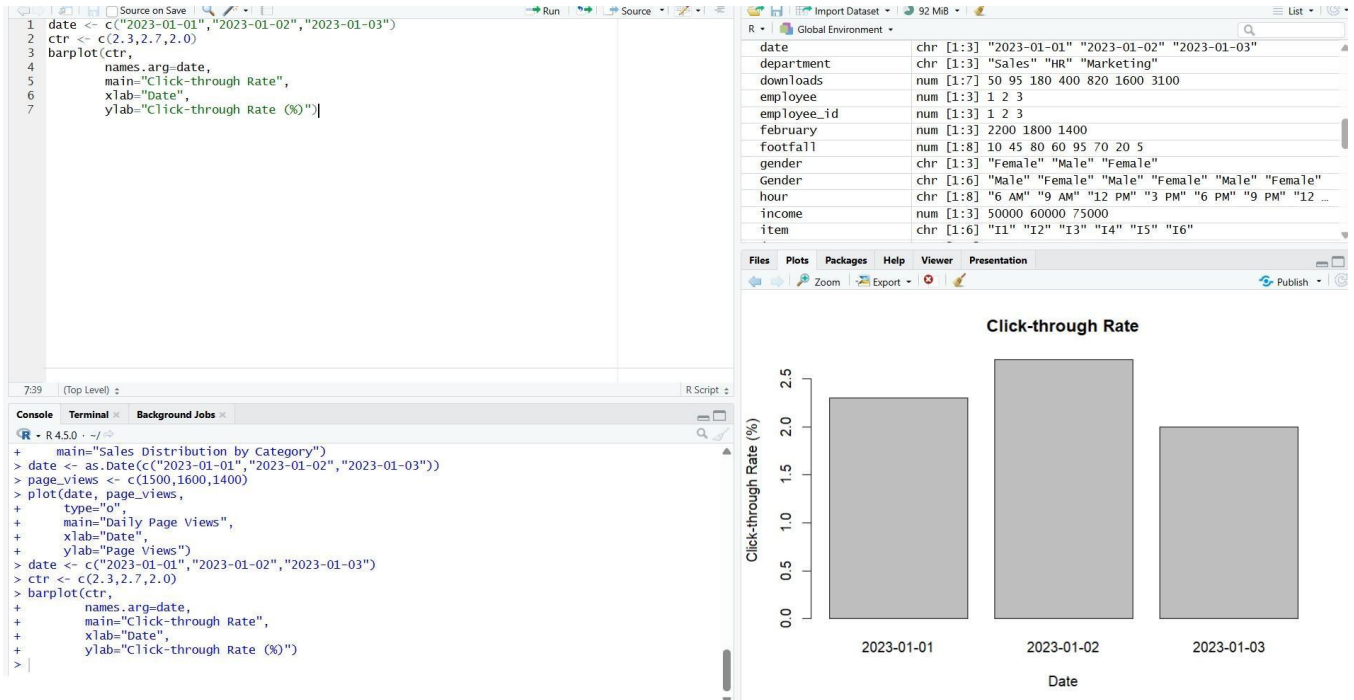


Result: Page views rise from 1500 to a peak of 1600 on 2023-01-02, then fall to the lowest value of 1400 on 2023-01-03.

Question 2: Bar Chart of Top N Days by Click-through Rate

Question: Generate a bar chart showing the top N days with the highest click-through rates. Label the chart elements.

Aim: To identify the top N days with the highest click-through rates using a labelled bar chart.

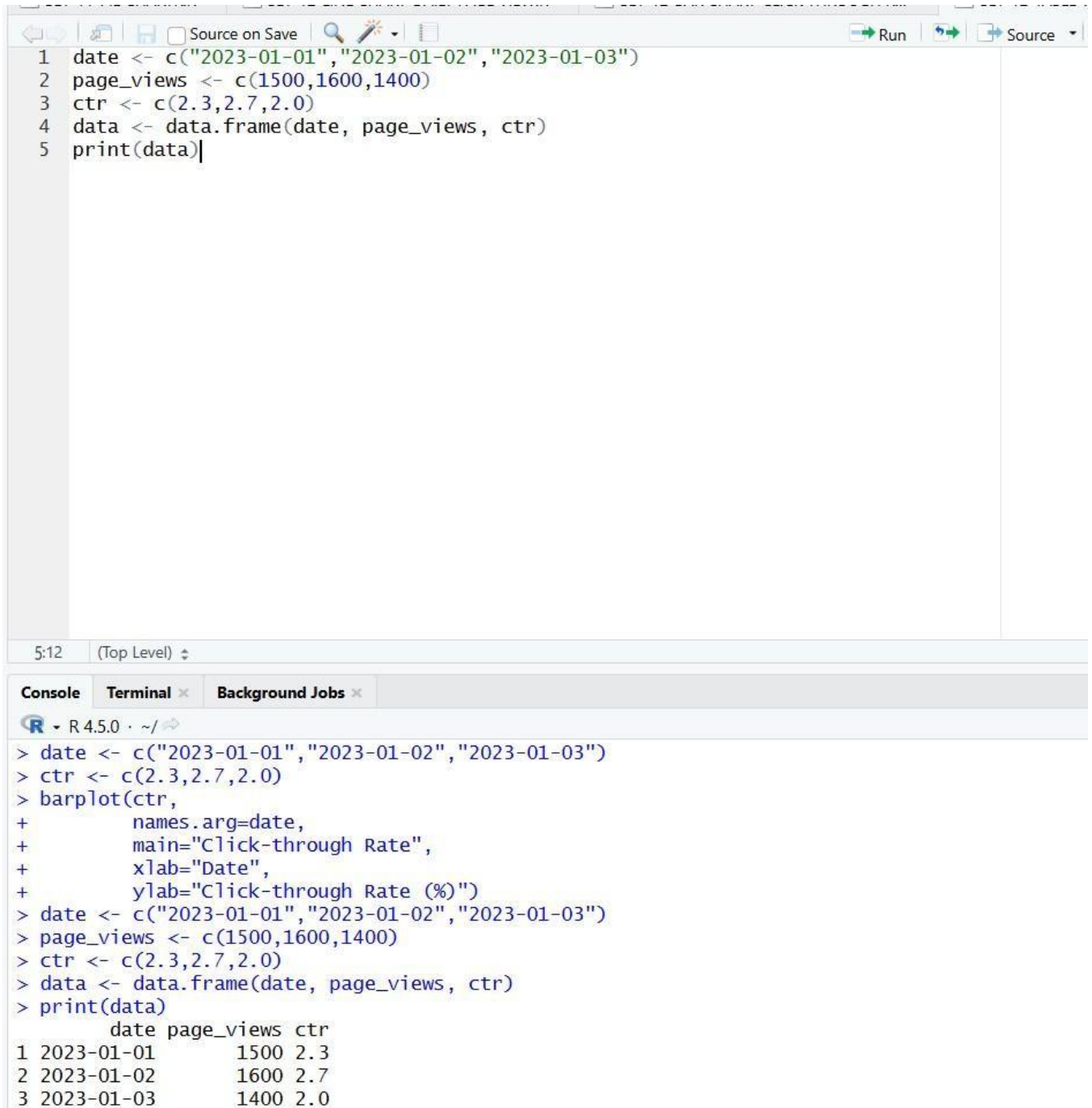


Result: For N = 2, 2023-01-02 ranks first (2.7%) and 2023-01-01 second (2.3%), while 2023-01-03 has the lowest rate (2.0%).

Question 3: Table of Website Traffic

Question: Build a table to show the website traffic data for each day. Label the table elements.

Aim: To present the page views and click-through rate of each day in a clearly labelled table.



The screenshot displays the RStudio environment. The top pane, labeled 'Source', contains the following R code:

```
1 date <- c("2023-01-01", "2023-01-02", "2023-01-03")
2 page_views <- c(1500, 1600, 1400)
3 ctr <- c(2.3, 2.7, 2.0)
4 data <- data.frame(date, page_views, ctr)
5 print(data)
```

The bottom pane, labeled 'Console', shows the execution of the code. It includes the same code as the Source pane, followed by the output of the `print(data)` command:

```
> date <- c("2023-01-01", "2023-01-02", "2023-01-03")
> ctr <- c(2.3, 2.7, 2.0)
> barplot(ctr,
+         names.arg=date,
+         main="Click-through Rate",
+         xlab="Date",
+         ylab="Click-through Rate (%)")
> date <- c("2023-01-01", "2023-01-02", "2023-01-03")
> page_views <- c(1500, 1600, 1400)
> ctr <- c(2.3, 2.7, 2.0)
> data <- data.frame(date, page_views, ctr)
> print(data)
      date page_views ctr
1 2023-01-01      1500 2.3
2 2023-01-02      1600 2.7
3 2023-01-03      1400 2.0
```

Result: The table lists the three days with their page views and click-through rates; 2023-01-02 is highest in both.

Question 4: Interactive Tableau Dashboard

Question: Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of website traffic data.

Aim: To build an interactive Tableau dashboard that combines the line chart, bar chart and table for exploring website traffic data.



Result: The dashboard places all three views together, and a date filter updates them so users can explore the traffic interactively.

Overall Result

Observation: The average page views are 1500 per day and the average click-through rate is about 2.3%, with 2023-01-02 the best day for both.